

BALAJI

PRODUCT REQUIREMENTS DOCUMENT

A high-trust, animated omnichannel electronics storefront for retail and wholesale buyers

Product: Balaji Electronic Commerce Experience

Version: 1.0 — proof-of-concept definition

Primary market: Pune & PCMC, Maharashtra

Platforms: Responsive web, mobile web, installable PWA

Audience: Founder, product, UX, engineering, merchandising, operations

NORTH STAR Help a Pune buyer move from “I need the right electronics product” to a confident order or quote in the fewest meaningful actions—without hiding price, availability, warranty, delivery, or wholesale terms.

Executive decision

Build one storefront with two task modes: Retail and Wholesale. The same inventory powers both, but pricing, quantity controls, trust cues, checkout, documents, and support paths adapt after lightweight role selection. Browsing remains open; sign-in is required only for saved state, protected B2B pricing, quotes, and checkout.

Success at a glance

Outcome	Proof-of-concept target	Guardrail
Findability	80% of test users reach a suitable product in ≤ 3 interactions	Search and category browse always visible
Retail conversion	$\geq 3.0\%$ checkout conversion in usability benchmark	No forced account before cart
Wholesale utility	Quote or bulk-cart completion $\geq 20\%$ of qualified sessions	GST and MOQ visible before commitment
Performance	LCP $\leq 2.5s$ p75 mobile; INP $\leq 200ms$; CLS ≤ 0.1	3D never blocks product discovery
Trust	Warranty, seller, delivery and returns visible on PDP	No misleading scarcity or fake reviews

1. Product vision and scope

Problem

Electronics shopping creates high cognitive load: similar model numbers, technical specifications, uncertain compatibility, hidden installation costs, warranty anxiety, and price comparison. Wholesale buyers add MOQ, tier pricing, GST documentation, repeat ordering, approval, and delivery coordination. Most storefronts force both audiences through the same flow.

Vision

Balaji Electronic becomes Pune's most helpful digital electronics counter: visually premium, locally credible, unusually clear about money and fulfilment, and fast enough for practical daily use. Motion and 3D communicate product form and system state; they are not decorative delays.

In scope for POC

- Responsive landing, category, search, listing, comparison, product detail, cart, checkout, order confirmation and account surfaces.
- Retail/Wholesale mode selection and task-based authentication; protected wholesale price tiers and quote workflow.
- 500 seeded products across 20 categories; dummy prices, stock, media, specifications, ratings and fulfilment.
- Bookmarks/wishlist, recently viewed, comparison, saved carts, replenishment lists and shareable inventory QR.
- Site Guide Bot that explains navigation, products, compatibility, policies and current screen; safe handoff to human support.
- Payment-link-forward checkout, UPI/cards/net banking/wallets/COD configuration, EMI messaging, GST invoice, delivery slot and installation.
- Accessible, reduced-motion, low-data and low-power modes.

Explicitly out of scope for POC

- Real payment capture, ERP reconciliation, production courier booking and live tax filing.
- Scraping or republishing third-party seller images, prices or claims without permission.
- Native iOS/Android apps; PWA covers installability in phase one.
- Autonomous bot refunds, price commitments, or warranty adjudication.

Assumptions to validate

Balaji operates or intends to operate from Pune; final store address, legal entity name, GSTIN, warehouse coverage, service partners, brand authorizations, returns policy and payment provider are not yet supplied. All such fields must remain clearly marked placeholders until verified.

2. Market and audience insight

Pune market signal

The local scan shows broad demand spanning TVs, cooling, major appliances, kitchen products, mobiles, computing and accessories. Pune-based B2B distributors emphasize dealer registration, GST billing, bulk supply, protected pricing and relationship management. Computing specialists add components, custom PCs and peripherals. Community references repeatedly point to Budhwar Peth/Tapkir Galli for components; treat that as directional qualitative evidence, not a verified merchant directory.

PRODUCT IMPLICATION The home page must feel like a premium retailer, while wholesale tools must behave like a procurement console. The catalog taxonomy must accommodate both boxed consumer goods and compatibility-sensitive parts.

Primary personas and jobs

Persona	Job to be done	Decision cues	Fast path
Retail upgrader	Choose a trustworthy appliance or device for home	Final payable price, delivery, installation, warranty, exchange, EMI	Search → compare → PDP → Buy now
Value seeker	Find the best fit inside a budget	Deal clarity, price history label, reviews, alternatives	Budget chip → shortlist → compare
Retailer/dealer	Replenish multiple SKUs profitably	Dealer price, MOQ, slabs, stock, GST, ETA	Wholesale mode → quick order → quote/pay
SME/institution	Buy a configured bundle with approval	Formal quote, tax invoice, delivery plan, support SLA	Use-case bundle → quote → approval
Maker/technician	Find compatible parts quickly	Voltage, connector, dimensions, datasheet, pickup	Part search → compatibility → pickup

Psychology principles—ethical use

- Recognition over recall: persistent search, recent queries, visually distinct categories and remembered mode.
- Choice architecture: lead with use-case and budget filters; progressively disclose expert specifications.
- Hick's law: one dominant CTA per decision state; secondary actions visibly secondary.
- Fitts's law: 44×44 px minimum targets; mobile primary CTA in thumb-reachable sticky bottom bar.
- Peak-end effect: make delivery promise, payment confirmation and post-purchase next steps exceptionally clear.
- Loss aversion without manipulation: use verified inventory and offer expiry only when sourced from real systems.
- Trust transfer: show authorized-brand status only when documented; display warranty owner and installation responsibility.

3. Experience architecture

Global navigation

Area	Desktop	Mobile
Header	Logo, delivery location, large search, mode switch, account, bookmark, cart	Compact logo, search trigger, mode pill, bookmark, cart
Category access	Mega menu grouped by use case and department	Bottom-sheet category browser
Primary mobile nav	Not required	Home, Categories, Search, Saved, Account
Urgent utility	Running announcement rail with pause control	Single-line swipe rail with pause
Help	Floating Site Guide Bot; never covers CTA	Docked help orb above sticky CTA

Minimum-click task routes

1. Known product: focus search on load → autocomplete → product → Buy now → payment.
2. Exploratory retail: use-case tile → pre-filtered listing → compare up to 4 → product/cart.
3. Wholesale repeat: sign in → Repeat order → edit quantities → pay or request quote.
4. Inventory sharing: tap Share inventory → scope selection → QR/link → recipient opens live filtered catalog.
5. Support: Site Guide suggests the next action based on current page and preserves context during human handoff.

Information architecture

Home → department/category → search/listing → compare → product → cart → checkout → payment/result → order tracking. Account contains profile, addresses, GST/business profile, orders, quotes, invoices, bookmarks, saved carts, replenishment lists, alerts and support. Content includes buying guides, compatibility explainers, warranty/returns, installation and Pune delivery coverage.

Retail vs wholesale behavior

Capability	Retail mode	Wholesale mode
Entry	Browse immediately; optional sign-in	Browse catalog; verify business to reveal tiers
Price	MRP, offer, total savings, EMI	Base/dealer price, GST state, slab tiers, MOQ
Cart	Unit quantity, delivery/install add-ons	Case quantity, multi-SKU grid, upload/order by SKU
Checkout	Address, slot, payment, invoice	PO/quote, GST, credit terms if approved, multi-address
Saved state	Wishlist/bookmarks and alerts	Lists, templates, repeat orders and account pricing

4. Landing page specification

Narrative sequence

Zone	Purpose and content	Primary action / behavior
1. Utility rail	Pune delivery promise, store hours, verified offer; horizontally running	Pause; select delivery area
2. Hero	3D product constellation around a clear value proposition; Retail/Wholesale task cards	Shop retail / Buy wholesale
3. Smart search	Natural-language prompt: “43-inch TV under ₹35k with installation”	Search with voice/text and recent queries
4. Category orbit	12 priority categories, 8 secondary; tactile cards with depth	Open pre-filtered category
5. Deal theatre	One genuine campaign with price, end time and stock source	View offer; no carousel auto-rotation
6. Choose-by-need	New home, office setup, gaming, cooling, creator, repair/maker bundles	Start guided bundle
7. Compare strip	Popular comparison trios with plain-language differences	Compare
8. Pune trust	Delivery coverage, pickup, installation, GST invoice, support proof	Check PIN code / contact
9. Wholesale desk	MOQ, price tiers, quote turnaround, dealer onboarding	Create business account
10. Social proof	Verified purchase reviews and project stories	Read proof
11. Inventory QR	Live catalog-sharing preview and scope controls	Generate QR
12. Footer	Policies, categories, brands, service, business identity	Direct navigation

Hero interaction rules

- Load a static poster immediately; hydrate the lightweight 3D scene after primary content.
- Pointer parallax ≤ 6 px and object rotation $\leq 8^\circ$; no continuous motion when the user is reading.
- Retail and Wholesale cards explain the consequence before selection. Remember mode, but keep switch visible.
- Hero CTA remains text-first and high contrast. 3D objects never become the only navigation affordance.
- On reduced-motion, replace orbit/scroll choreography with fades ≤ 150 ms and a static product composition.

Visual direction

Premium Pune-tech aesthetic: deep navy/ink base, electric blue and cyan for technology, restrained saffron/orange for commerce and local warmth, translucent glass panels over controlled gradients, crisp product photography, generous dark-to-light transitions and rounded geometry. Glass surfaces must retain opaque fallback and at least 4.5:1 text contrast. Use the Dribbble reference for editorial composition, large product imagery and whitespace—not for literal copying.

5. Core journeys and functional requirements

Authentication and roles

- Start with Retail / Wholesale intent; allow guest retail browsing and cart.
- Retail sign-in: mobile OTP primary; email/password optional; social sign-in only if operationally supported.
- Wholesale onboarding: mobile/email OTP, legal/business name, GSTIN where applicable, address, buyer role, expected monthly volume and consent.
- States: guest, retail customer, wholesale pending, wholesale verified, wholesale suspended, staff/admin.
- Step-up verification for sensitive changes, saved payment instruments, credit terms and high-value orders.
- Account merge and role switching must preserve bookmarks and carts without exposing wholesale pricing to unauthorized users.

Search, discovery and comparison

- Typo-tolerant, synonym-aware search across title, brand, model, SKU, category, attributes and compatibility.
- Autocomplete groups products, categories, brands, guides and recent queries.
- Facets update counts without page reload; selected filters become removable chips; mobile filter is a full-height sheet.
- Sort: relevance, price, rating, newest, discount, availability and wholesale margin where authorized.
- Compare 2–4 products with sticky difference-only toggle and plain-language “best for” summary.
- Zero results offers correction, adjacent category, remove-filter options and human help.

Product detail page

- Above fold: gallery/3D, title/model, rating, final price or protected B2B price, savings, stock, PIN-code ETA, warranty and dominant CTA.
- Variant choices show price/stock consequence instantly; unavailable combinations are disabled with reason.
- Gallery supports image, short video, 360/3D and dimension overlay; thumbnails remain keyboard accessible.
- Compatibility block answers “works with...” and highlights required accessories.
- Sticky mobile buy bar: price, Add to cart and Buy now/payment link; bookmark icon near title/gallery.
- Below fold: highlights, specs, comparison, offers, delivery/install, reviews, Q&A, documents and alternatives.

Cart, checkout and payment

- Persistent cart across devices after sign-in; clear stock reservation semantics.
- Cart shows complete payable estimate, delivery/install, GST state, coupon and savings; no surprise fee at final step.
- Express “Buy now” skips cart for a single eligible item.
- Checkout uses one page with collapsible completed sections: contact → delivery → invoice → payment.
- Payment options: UPI intent/QR, cards, net banking, wallets, configured COD, EMI and a prominent secure payment link.
- Payment link is copyable, revocable, time-bound, amount-locked, order-bound and status-synced; warn before sharing.
- Failure state preserves cart/order and offers Retry, Change method, or Get help. Never create duplicate orders on retries.

Bookmarks and saved work

A bookmark is available on every product card and PDP. Guests store locally and are prompted to sync after sign-in. Users can create lists, add notes, share read-only lists, enable price/stock alerts, move items to cart and compare saved items. Wholesale users can convert a list to a repeat-order template or quote request.

6. Site Guide Bot and inventory QR

Site Guide Bot

The bot's primary role is wayfinding and explanation: "What can I do here?", "Show wholesale prices", "How does installation work?", "Compare these TVs", or "Where is my order?". It may summarize approved catalog and policy content, invoke deterministic site actions with confirmation, and hand off context to support.

Bot capability	Allowed behavior	Safety boundary
Site tour	Contextual guided steps; user-controlled next/skip	Never trap focus or auto-start audio
Product help	Grounded summary from structured attributes and approved guides	No invented compatibility, stock or warranty
Navigation	Open search, filters, saved items, cart, account page	Confirm before destructive or financial action
Order help	Read authorized order status and explain next step	Authenticate; redact sensitive information
Commerce	Prepare cart/quote and deep-link to checkout	Cannot place order or promise price autonomously
Escalation	Create support ticket/chat handoff with transcript consent	Clear bot identity and easy dismissal

Bot UX requirements

- Visible "AI site guide" label; never impersonate a human.
- Suggested prompts tied to the page; free text always available.
- Citations/deep links to product specs and policy sections inside responses.
- Conversation can be cleared; transcript retention and consent are explicit.
- Marathi, Hindi and English language architecture; POC can launch English first with i18n-ready copy.
- Fallback uses deterministic help search when model confidence or retrieval quality is low.

Share entire inventory by QR

The default QR should encode a short HTTPS URL to a live inventory landing page, not 500 product records. This keeps the QR scannable and inventory current. Authorized users choose scope: full public catalog, category, brand, saved list, quote, or wholesale catalog. Wholesale links require authentication before protected pricing is shown.

- QR preview, downloadable PNG/SVG, copy link, native share and print label.
- Link metadata stores creator, scope, filters, expiry, campaign and revocation state.
- Public page uses canonical URL, no sensitive query parameters, Open Graph preview and analytics consent.
- QR error correction level M by default; minimum printed size tested at intended distance.
- Track generated → opened → product viewed → cart/quote, without exposing personal data.

7. Motion, 3D and micro-interactions

Motion system

Interaction	Motion	Timing / constraint
Button hover	2 px lift, subtle highlight sweep, icon translate	120–180 ms; no layout shift
Button press	Scale to 0.98, shadow contracts	80–120 ms; immediate state feedback
Bookmark	Icon fill + tiny radial pulse	180–240 ms; announce saved state
Add to cart	Product thumbnail arcs toward cart only on capable devices	≤450 ms; instant count update
Filter apply	Result skeleton/crossfade; chip settles	150–220 ms; preserve scroll
Page transition	Shared element for product image; short fade otherwise	≤300 ms; cancellable
Running rail	Constant linear scroll with pause on hover/focus	No critical info exclusively animated
3D viewer	Drag rotate, pinch zoom, hotspot focus	User-driven; cap DPR and frame budget

Performance tiers

- Tier A: static responsive image and CSS micro-interactions for all devices.
- Tier B: short product video/360 sprite after consent or interaction.
- Tier C: WebGL/glTF 3D only on capable devices and only after primary content.
- Stop offscreen animation; respect save-data, reduced-motion, thermal/battery signals where available.
- Maintain identical commerce functionality without WebGL, blur, hover or animation.

Accessibility acceptance

- Target WCAG 2.2 AA; keyboard-complete; logical heading and landmark structure.
- Visible focus, skip link, semantic controls, form labels, inline errors and error summary.
- Touch targets ≥44×44 px; text zoom to 200%; reflow at 320 CSS px without loss.
- Contrast: 4.5:1 normal text, 3:1 large text and UI boundaries. Glass does not reduce readability.
- Reduced motion disables parallax, auto-orbits, celebratory motion and smooth-scrolling dependence.
- Product imagery has meaningful alt text; decorative 3D is hidden from assistive tech; viewer has equivalent specs.

8. Catalog proof of concept

500-SKU composition

Category	Seed SKUs	Hero media
Mobiles	32	3D/360 for top 2-4; image for remainder
Laptops & tablets	34	3D/360 for top 2-4; image for remainder
Desktop & components	36	3D/360 for top 2-4; image for remainder
TV & home theatre	32	3D/360 for top 2-4; image for remainder
Audio	32	3D/360 for top 2-4; image for remainder
Refrigerators	24	3D/360 for top 2-4; image for remainder
Washing machines	24	3D/360 for top 2-4; image for remainder
Air conditioners & coolers	26	3D/360 for top 2-4; image for remainder
Kitchen appliances	32	3D/360 for top 2-4; image for remainder
Small home appliances	28	3D/360 for top 2-4; image for remainder
Cameras & imaging	16	3D/360 for top 2-4; image for remainder
Gaming	24	3D/360 for top 2-4; image for remainder
Networking	24	3D/360 for top 2-4; image for remainder
CCTV & security	22	3D/360 for top 2-4; image for remainder
Power & backup	24	3D/360 for top 2-4; image for remainder
Smart home	20	3D/360 for top 2-4; image for remainder
Wearables	18	3D/360 for top 2-4; image for remainder
Mobile accessories	32	3D/360 for top 2-4; image for remainder
Computer accessories	28	3D/360 for top 2-4; image for remainder
Electronic components & tools	24	3D/360 for top 2-4; image for remainder

Total: 500 seeded SKUs. The accompanying catalog CSV defines every item, indicative INR price, retail/wholesale eligibility, image direction and a lightweight animation treatment. Names and commercial values are synthetic; brand-like examples must be replaced or licensed before publication.

Product data completeness

- Required: SKU, title, slug, category, brand, model, price, tax class, stock, images, short description, key attributes, warranty, fulfilment flags and status.
- Wholesale: MOQ, pack size, tier breaks, protected price, lead time and quote eligibility.
- Media: 1:1 card image, 4:3 gallery, mobile crop, alt text, focal point, rights owner, license and expiry.
- 3D: glTF/GLB, Draco/KTX2 where appropriate, poster, dimensions, hotspots and fallback.
- Quality gate: no published product without price state, stock state, at least one licensed image, alt text and warranty owner.

Image and animation sourcing

For the POC, use generated or explicitly licensed images, or manufacturer media with documented permission. Do not hotlink search-result images. Product animation should be authored as reusable treatments—parallax card, hotspot reveal, 360 spin, exploded view, color swap—not 500 heavyweight videos. The seed CSV assigns a treatment per SKU so the UI can demonstrate variety efficiently.

9. Measurement, operations and governance

Event taxonomy

Journey	Core events	Primary metric
Discovery	search_submitted, autocomplete_selected, filter_applied, category_opened	Search success; zero-result rate
Evaluation	pdp_viewed, gallery_used, compare_added, bookmark_toggled	PDP→cart; save rate
Commerce	cart_added, checkout_started, payment_link_created, payment_result	Conversion; payment success
Wholesale	business_signup, verification_state, tier_viewed, quote_requested	Qualified activation; quote rate
Help	bot_opened, prompt_used, answer_feedback, handoff_started	Self-serve resolution
Share	qr_created, inventory_link_opened, shared_session_converted	Share→commerce rate

Operational consoles

- Catalog/PIM: products, variants, attributes, media, documents, SEO and publication states.
- Inventory/pricing: location stock, safety stock, price lists, offers, tiers and audit history.
- Orders/quotes: payment state, fulfilment, invoices, refunds, quote versions and approvals.
- Content: landing modules, announcement rail, buying guides, bot knowledge and localization.
- Support: tickets, bot handoffs, order context and SLA queue.
- Access: least-privilege roles, admin MFA, approval for price/policy changes and immutable audit log.

POC acceptance criteria

- All primary journeys work at 360, 768, 1024 and 1440 px reference widths.
- Role switching and authentication states never leak protected pricing.
- 500 products load from seed data; search, facets, sort, pagination/infinite loading and compare work.
- Bookmarking persists for guest and signed-in users; merge behavior is tested.
- QR opens the intended inventory scope and honors expiry/revocation.
- Bot answers only from approved content, links to source sections and hands off on uncertainty.
- Checkout demonstrates final price and payment link clearly; no actual charge in POC.
- Automated accessibility, unit, integration and end-to-end checks pass; manual keyboard and screen-reader smoke test passes.
- Performance budgets are met on a mid-tier mobile test profile with 3D disabled and enabled separately.

Risks and mitigations

Risk	Impact	Mitigation
3D and glass overuse	Slow, illegible, distracting	Progressive enhancement, opaque fallback, performance budget
Pricing leakage	Commercial and trust harm	Server-side authorization; separate price lists; tests
Stale inventory	Failed orders and support burden	Source timestamps, reservations,

		reconciliation and truthful labels
Bot hallucination	Wrong compatibility/policy advice	Retrieval-only knowledge, structured tools, confidence fallback
Image rights	Legal/reputation risk	Rights metadata, licensed/generative POC assets, takedown workflow
Dark patterns	Short-term lift, long-term distrust	Verified scarcity, clear totals, easy cancellation and consent

10. Delivery plan and open decisions

Phased roadmap

Phase	Outcome	Indicative scope
0 – Discovery	Validated roles, taxonomy and operating constraints	Stakeholder workshop, content audit, buyer interviews, service map
1 – UX prototype	Testable responsive experience	Design system, landing, search, PLP, PDP, role flows, checkout, bot prototype
2 – POC build	End-to-end demonstrator with 500 products	Seed catalog, auth sandbox, bookmarks, compare, QR, payment sandbox, analytics
3 – Pilot	Controlled Pune launch	Real catalog feeds, payments, fulfilment, support, security and performance hardening
4 – Scale	Operational optimization	ERP/PIM integration, personalization, multilingual content, advanced B2B

Open decisions before production

- Exact Balaji legal/store identity, address, service areas and fulfilment model.
- Consumer electronics only vs inclusion of components, repair tools and electrical goods.
- Owned stock, marketplace, dropship or hybrid inventory authority.
- Wholesale eligibility rules, price tiers, MOQ, credit policy and verification SLA.
- Payment provider, COD policy, EMI partners, cancellation/refund policy and payment-link expiry.
- Installation partners, warranty ownership, returns categories and reverse-logistics responsibility.
- Brand assets, Marathi/Hindi launch requirement, photography rights and final product feed.

Research sources

Maharashtra Electronics Corporation — Pune footprint and product range: <https://mymec.in/about-us/>

Raj Electronics Pune — local categories and delivery/installation cues: <https://www.rajelectronicspune.com/>

VIANET — Pune B2B distribution, GST billing and dealer pricing: <https://www.vianet.co.in/>

LeeComp Pune — wholesale computing and component mix: <https://www.leecomp.in/>

Bajaj Finserv/Croma Pune — mainstream category and store-location overview: <https://www.bajajfinserv.in/shop-on-emi-at-croma-store-in-pune>

Dribbble reference — Crookeries Ecommerce Website: <https://dribbble.com/shots/26163743-Crookeries-Ecommerce-Website>